Basic Science Department

BAS 141
Engineering Chemistry

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Chapter 2: Thermodynamics

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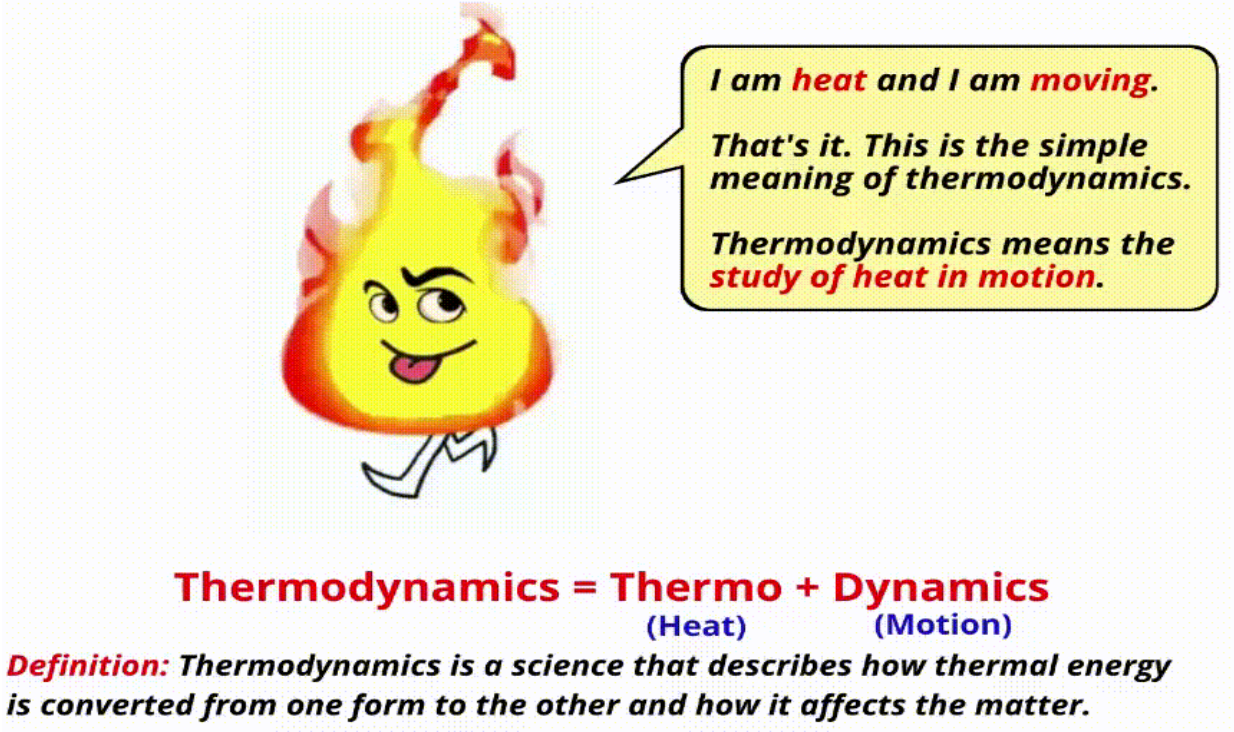
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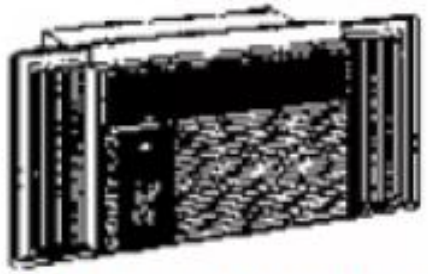
1.1. What is thermodynamics?

It studies energy, energy transformations, and its relation to matter.

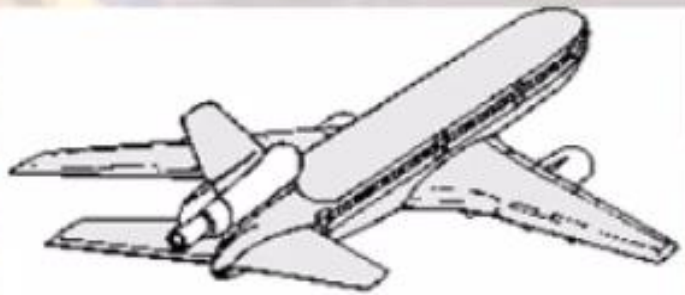


The study of thermodynamics concerns how energy is stored within a body and how energy transformations, which involve heat and work, may occur.

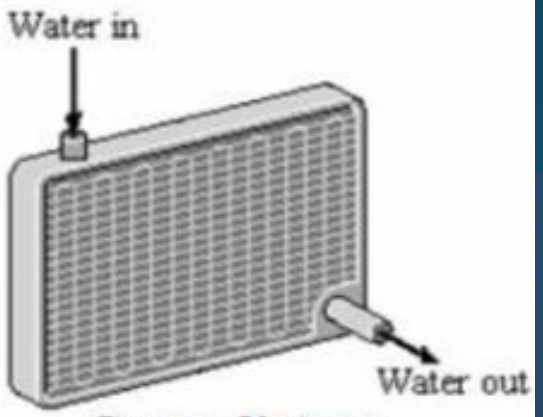
Applications of Thermodynamics



Air-conditioning systems



Airplanes



Car radiators



The human body

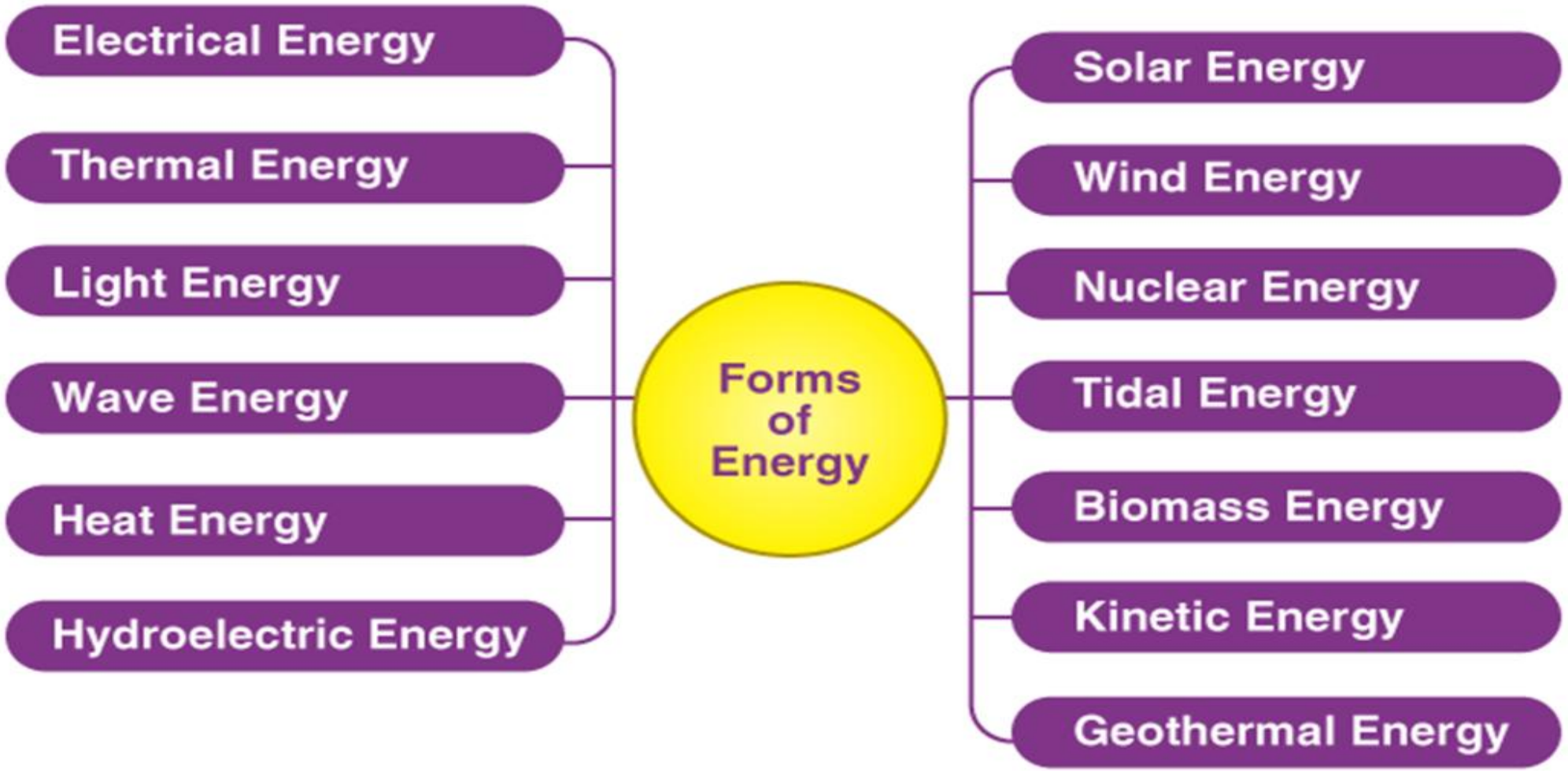


Power plants



Refrigeration systems

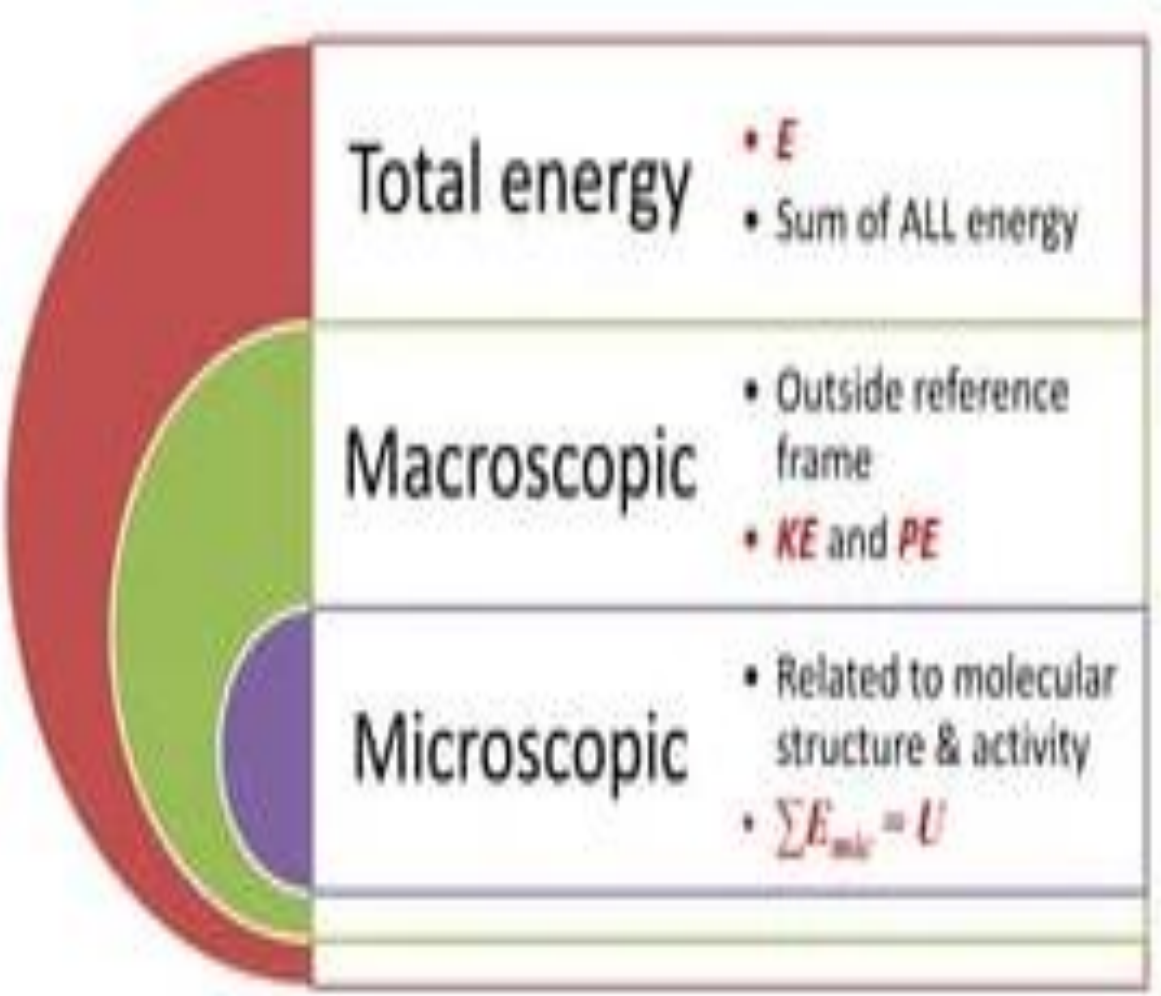
Energy is defined as the capacity to supply heat or do work.

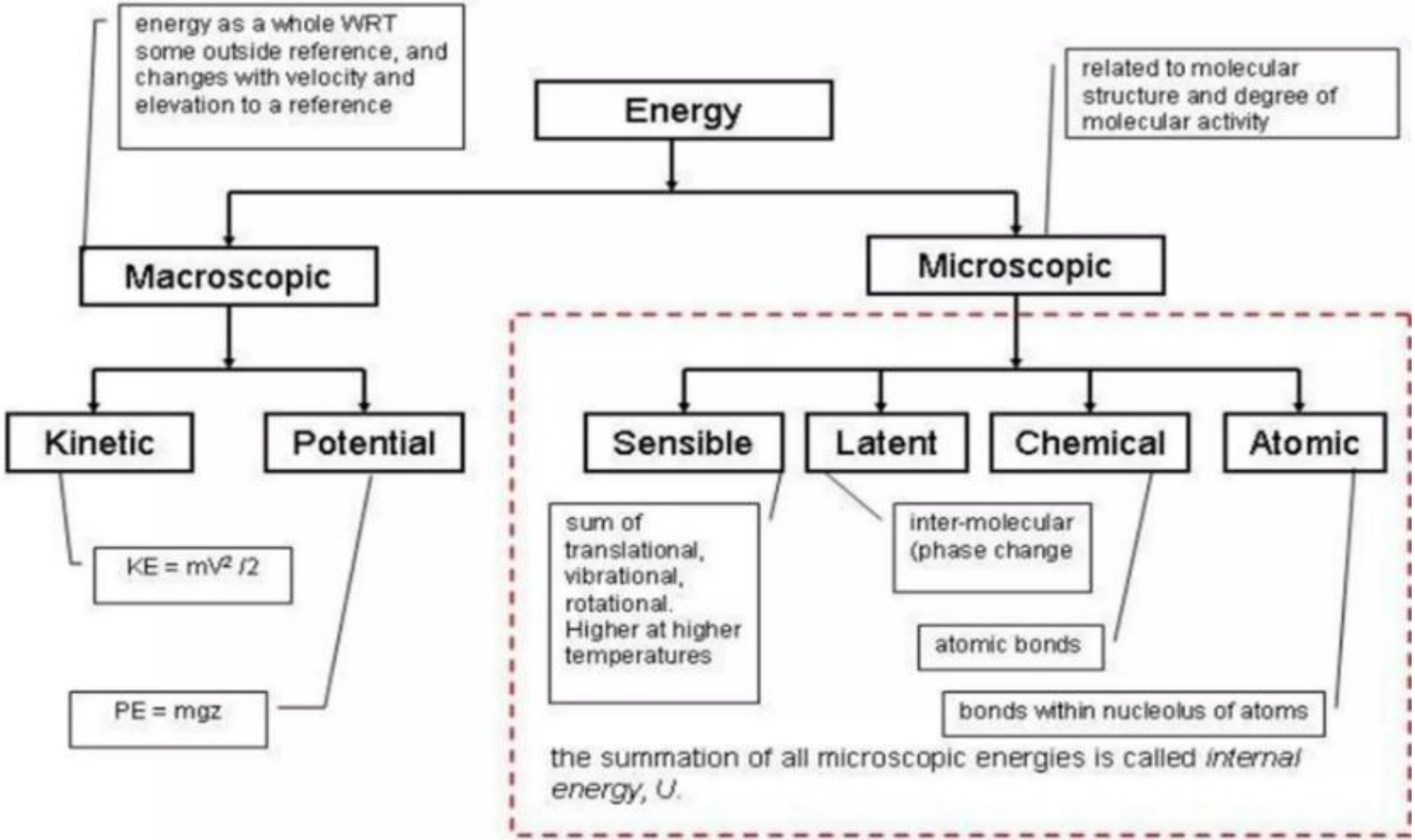


Approaches to studying thermodynamics

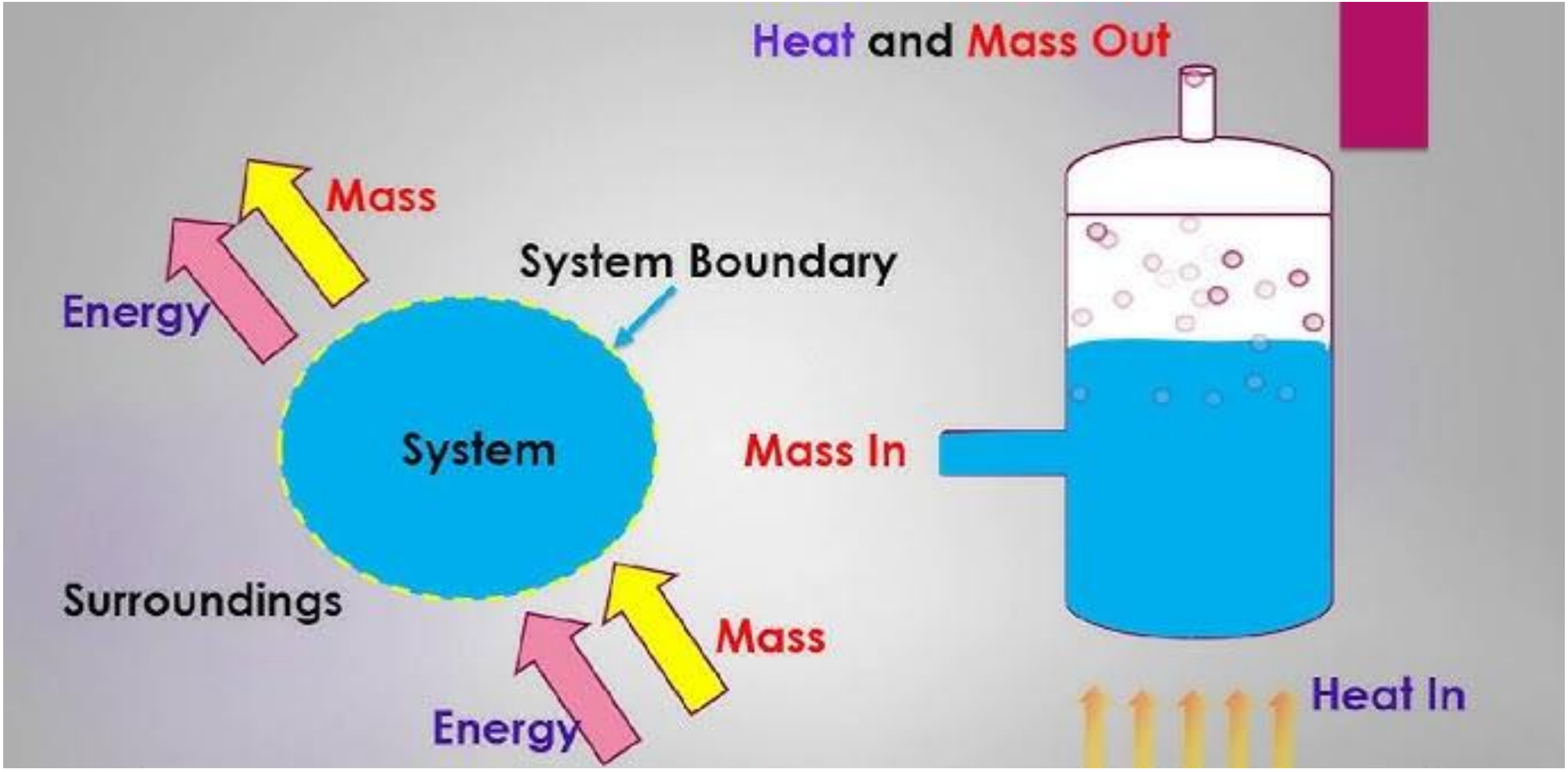
Macroscopic forms of energy are those that a whole system possesses concerning a fixed external reference.

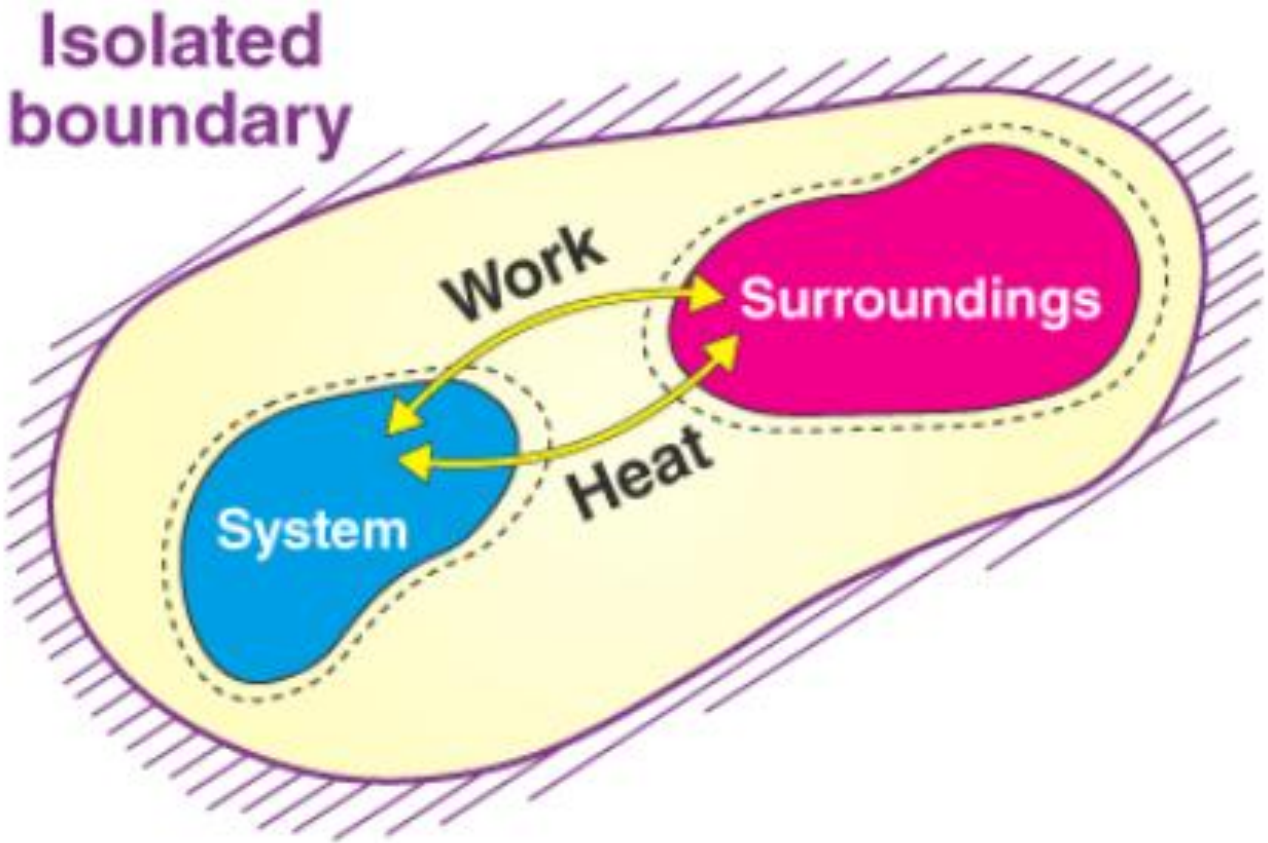
Microscopic forms of energy are those that relate to the system on a molecular or atomic level.




energy as a whole WRT some outside reference, and changes with velocity and elevation to a reference
Energy
related to molecular structure and degree of molecular activity
Macroscopic
Microscopic
Kinetic
Potential
Sensible
Latent
Chemical
Atomic
KE = mV² /2
PE = mgz
sum of translational, vibrational, rotational. Higher at higher temperatures
inter-molecular (phase change
atomic bonds
bonds within nucleolus of atoms
the summation of all microscopic energies is called *internal energy, U.*

1.2. Thermodynamic systems and their types





System boundary

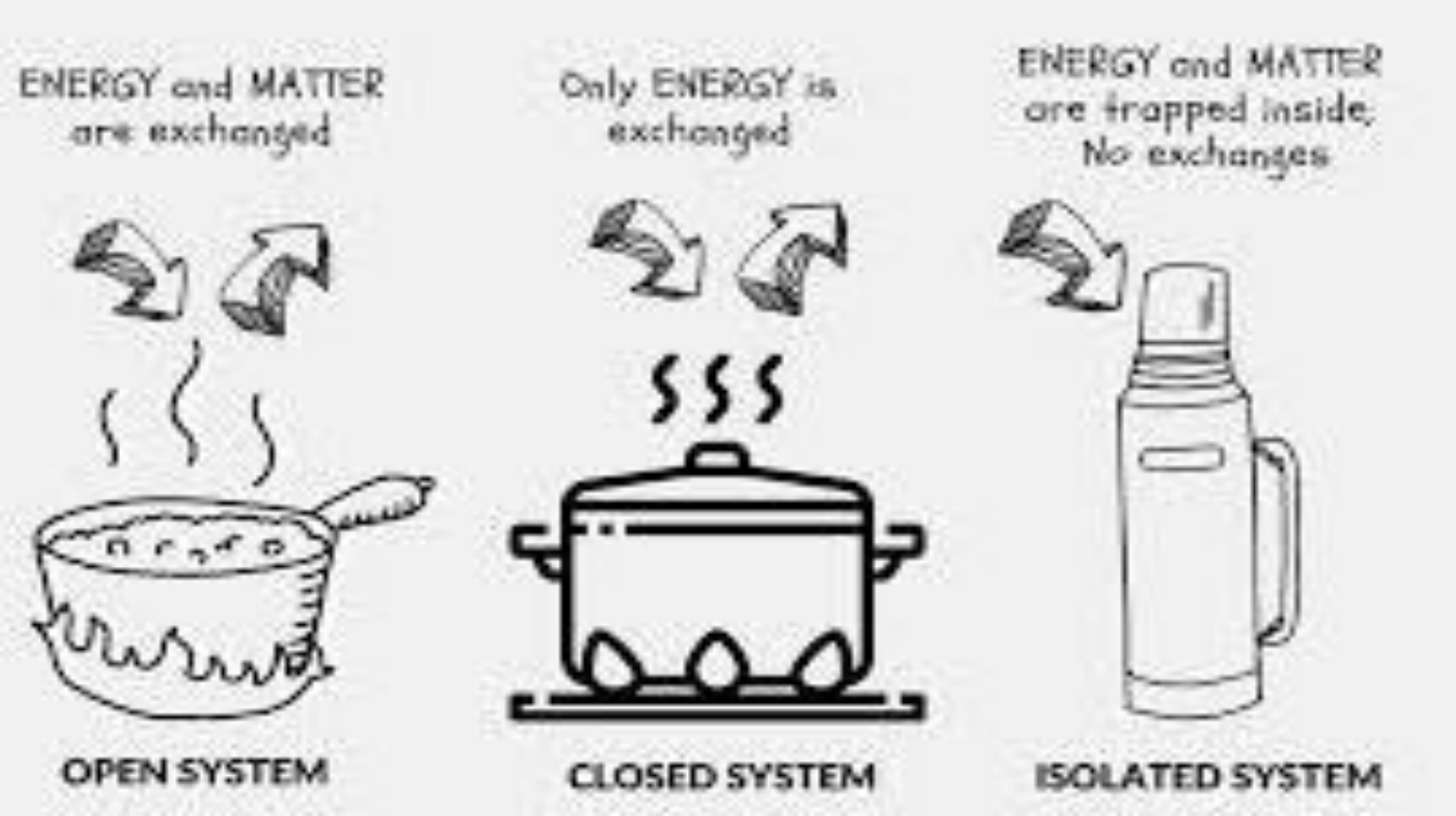
A boundary is a closed surface surrounding a system through which energy and mass may enter or leave the system.

Surroundings

Everything that interacts with the system.

System

A system is a region containing energy and/or matter that is separated from its surroundings by arbitrarily imposed walls or boundaries.



Systems types

A. Open System

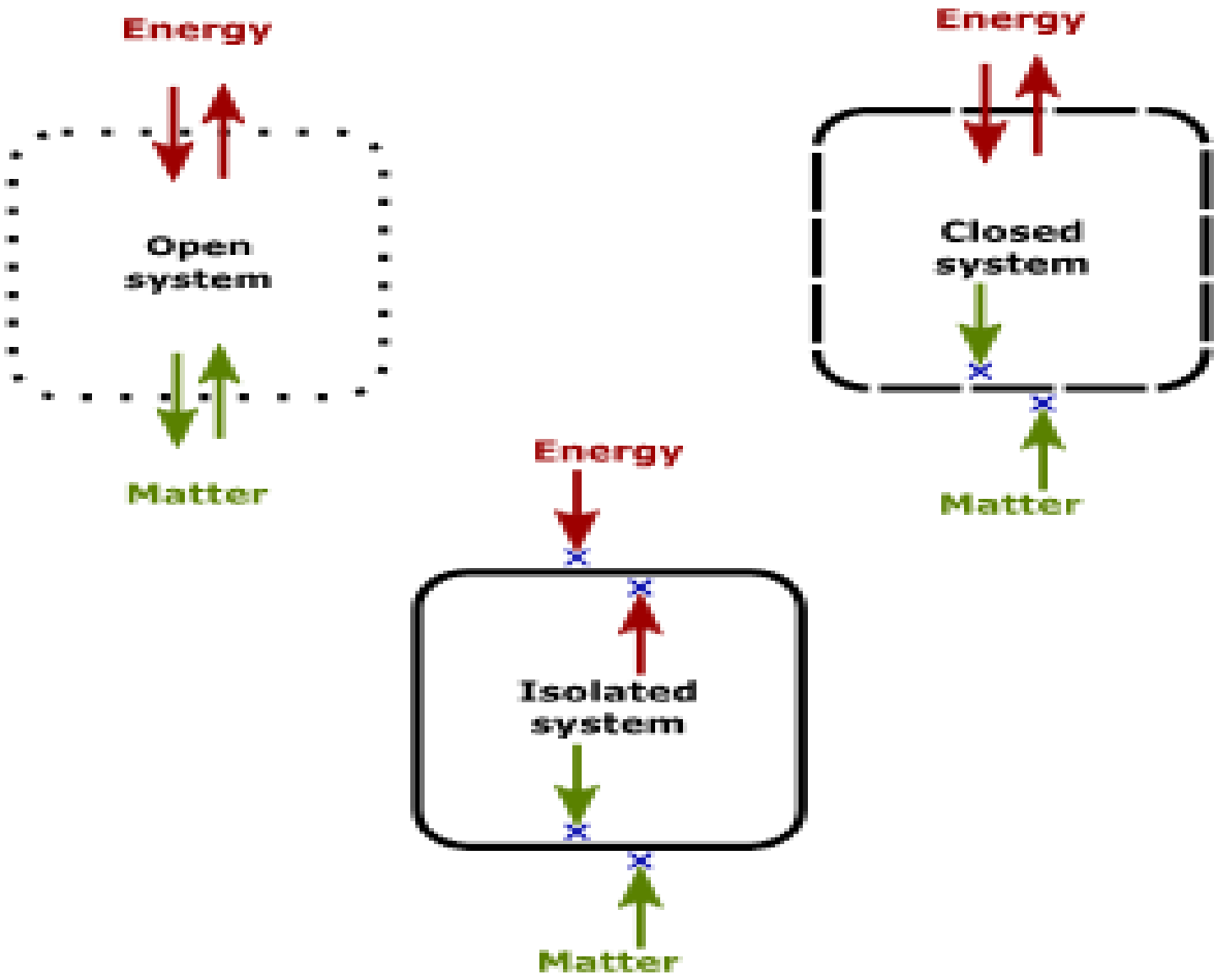
Can exchange matter and energy with surrounding.

B. Closed System

Exchange only energy with surrounding
(Can not exchange matter)

C. Isolated System

Neither matter nor energy can be exchanged with its surroundings.


Energy
Open system
Matter
Energy
Closed system
Matter
Energy
Isolated system
Matter

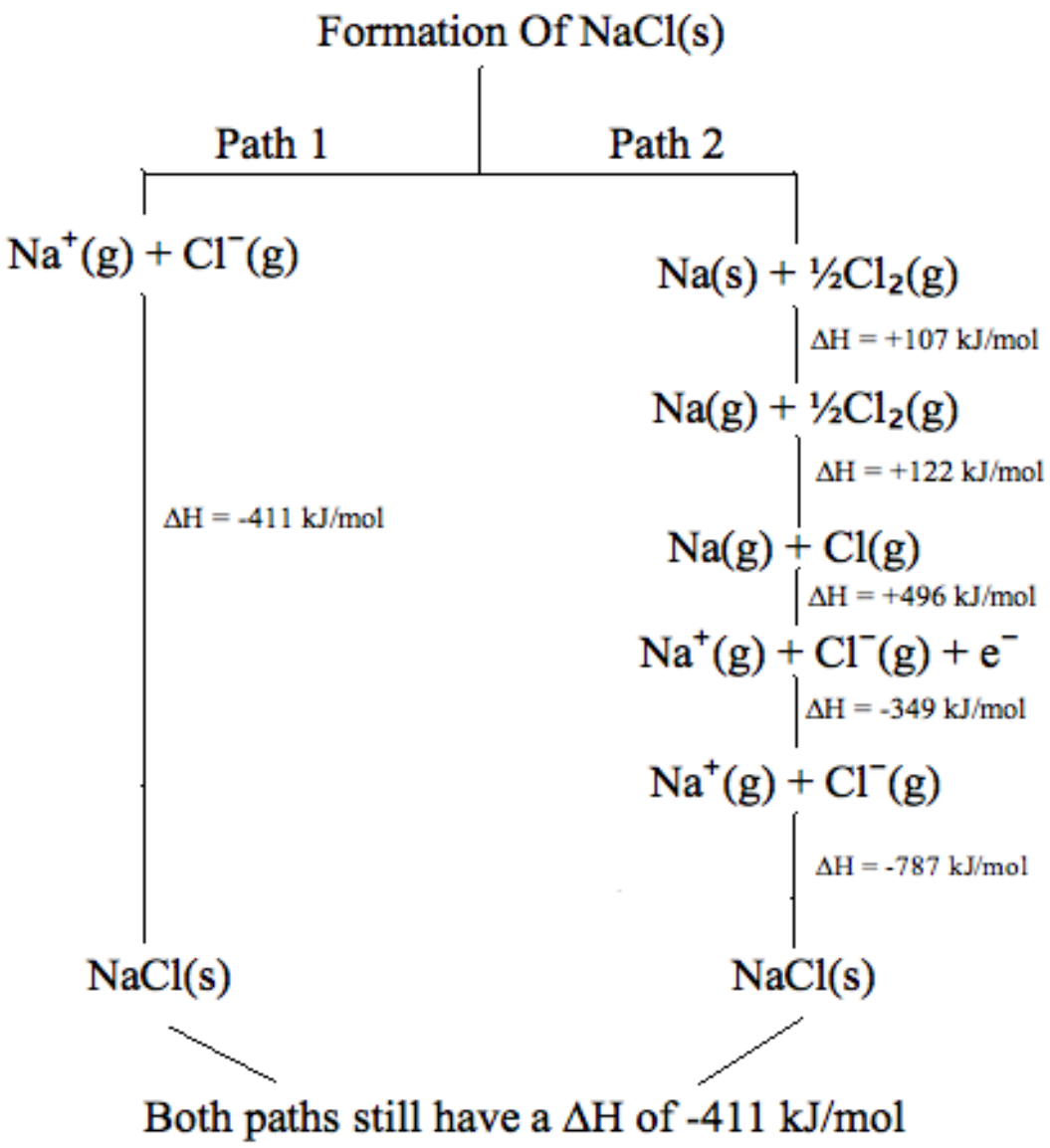
1.3. State Functions vs Path Functions

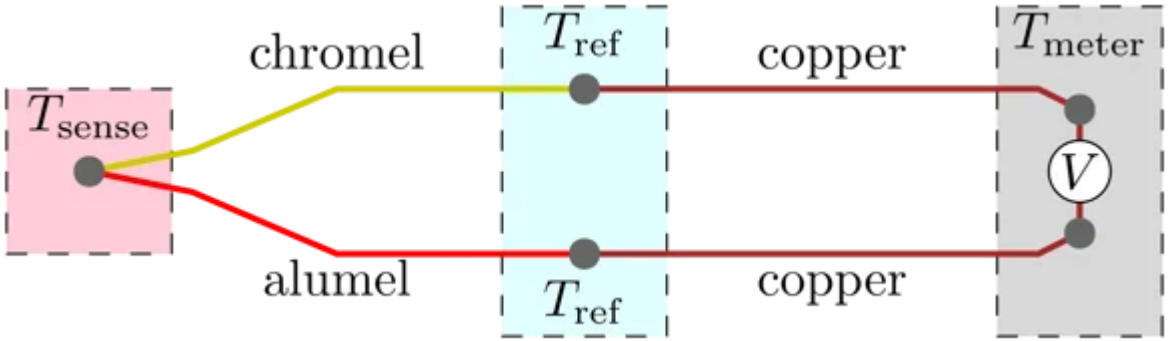
A **state function** is a property whose value **does not depend on the path** to reach that specific function or value.

State functions	Non state functions
E (Internal Energy)	q_b (heat)
S (Entropy)	w (work)
H (Enthalpy)	
P (Pressure)	
V (Volume)	
T (temperature)	

Path function **depends on the path** taken to reach that specific value.

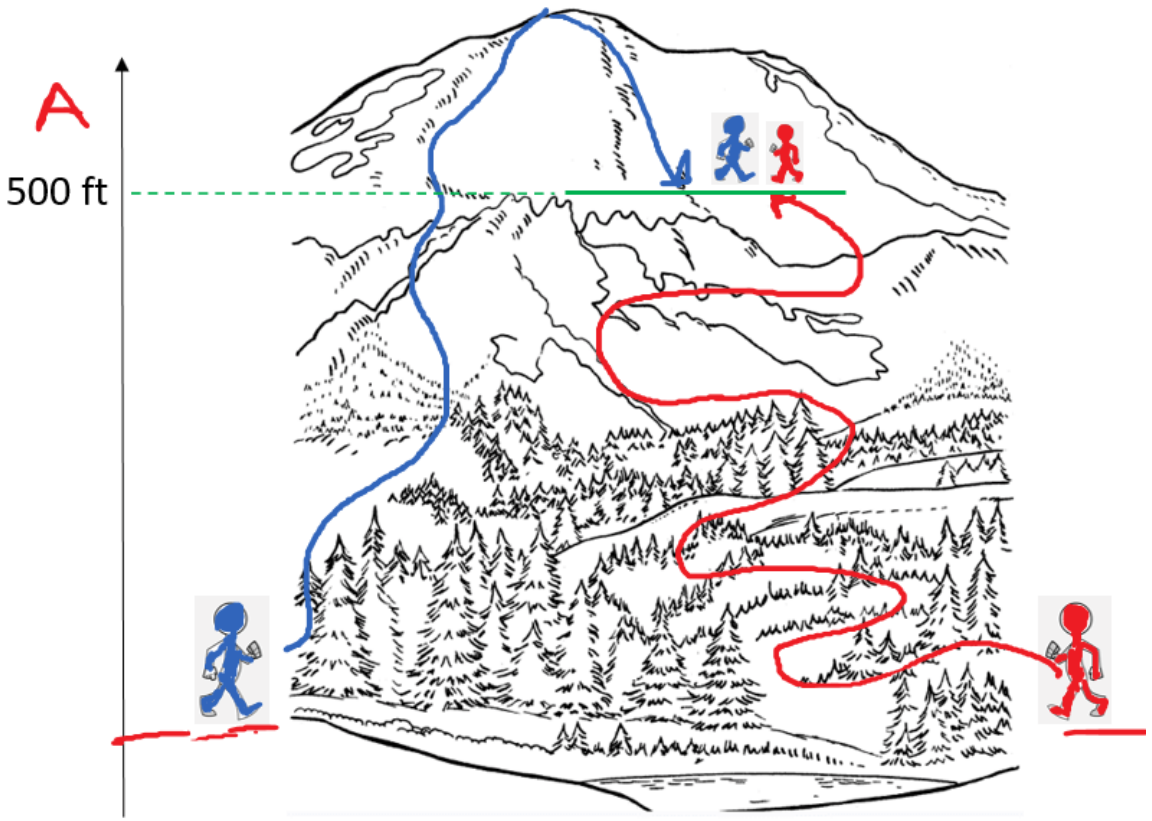
State function examples



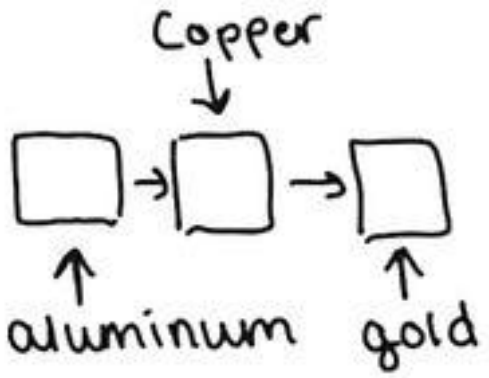


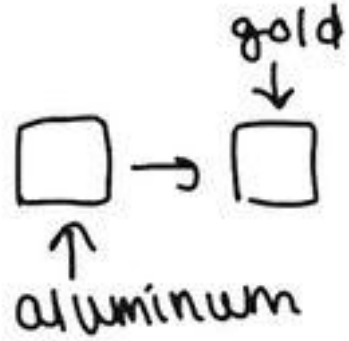
Path function examples

$$W = F.d$$



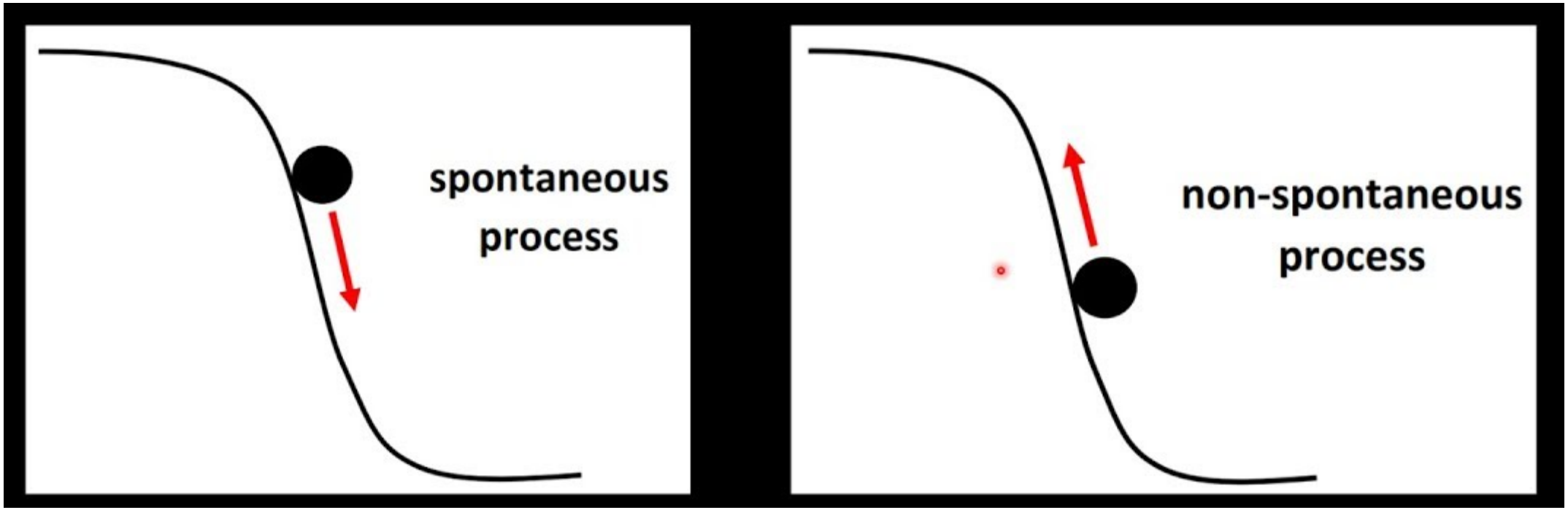
Heat





1.4. Spontaneous process

The spontaneous process proceeds on its own **without** any **external influence**



The **reverse** of a spontaneous process is always **nonspontaneous** and takes place only in the presence of some continuous external influence.

SPONTANEOUS PROCESSES: Characteristics of Spontaneous Processes

A **spontaneous process** is a physical or chemical change that occurs by itself. These processes occur without requiring an outside force and continue until equilibrium is reached.

EXAMPLES:



Ice melting at room temperature

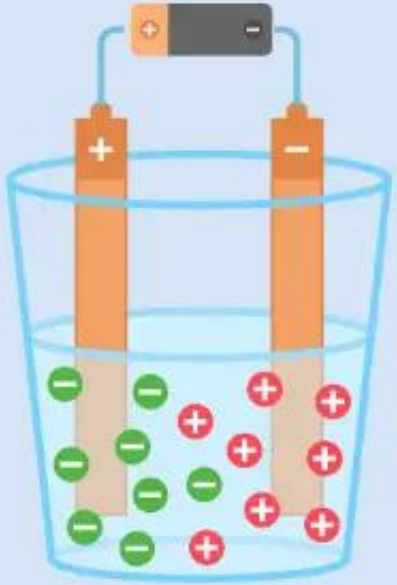


Iron rusts in moist air



Sugar dissolves in water

Examples of Non Spontaneous Reactions



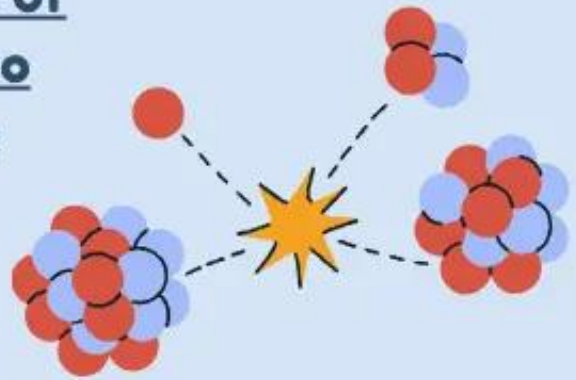
Electrolysis of water



Charging a battery



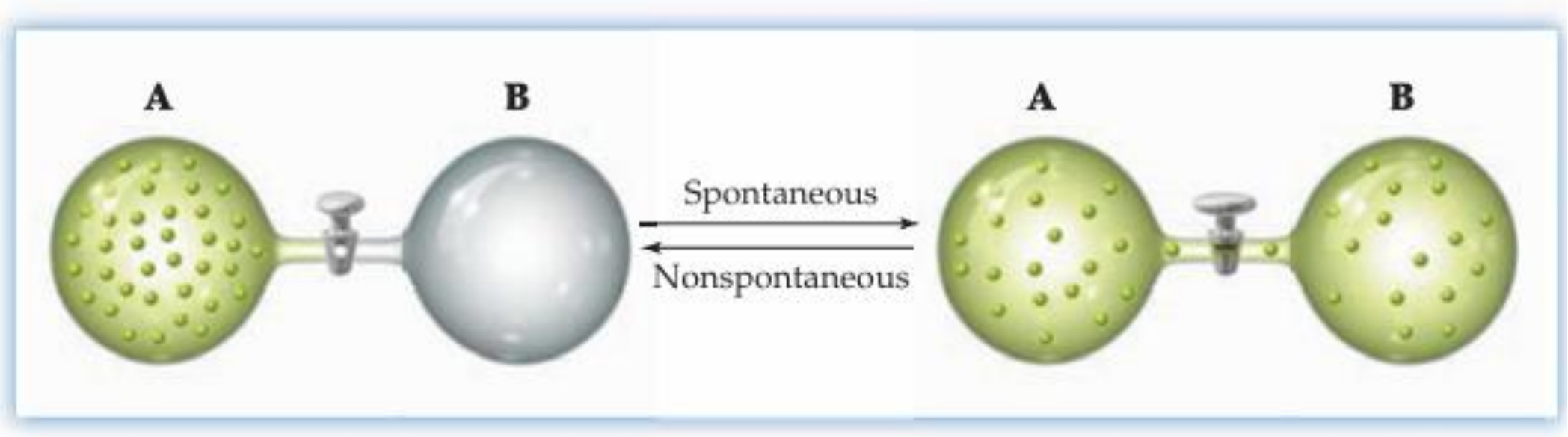
Conversion of diamond to graphite



Nuclear fusion



Decomposition of ozone into oxygen wood

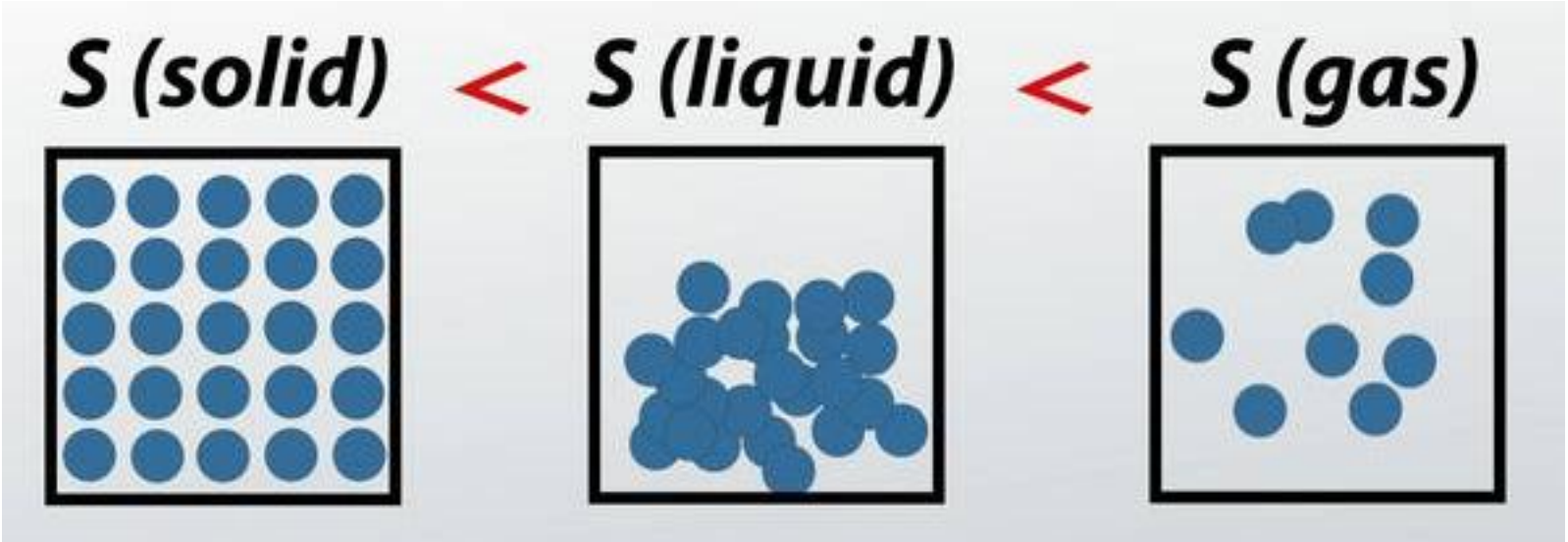


When the stopcock is opened, the gas in bulb A expands spontaneously into evacuated bulb B to fill all the available volume.

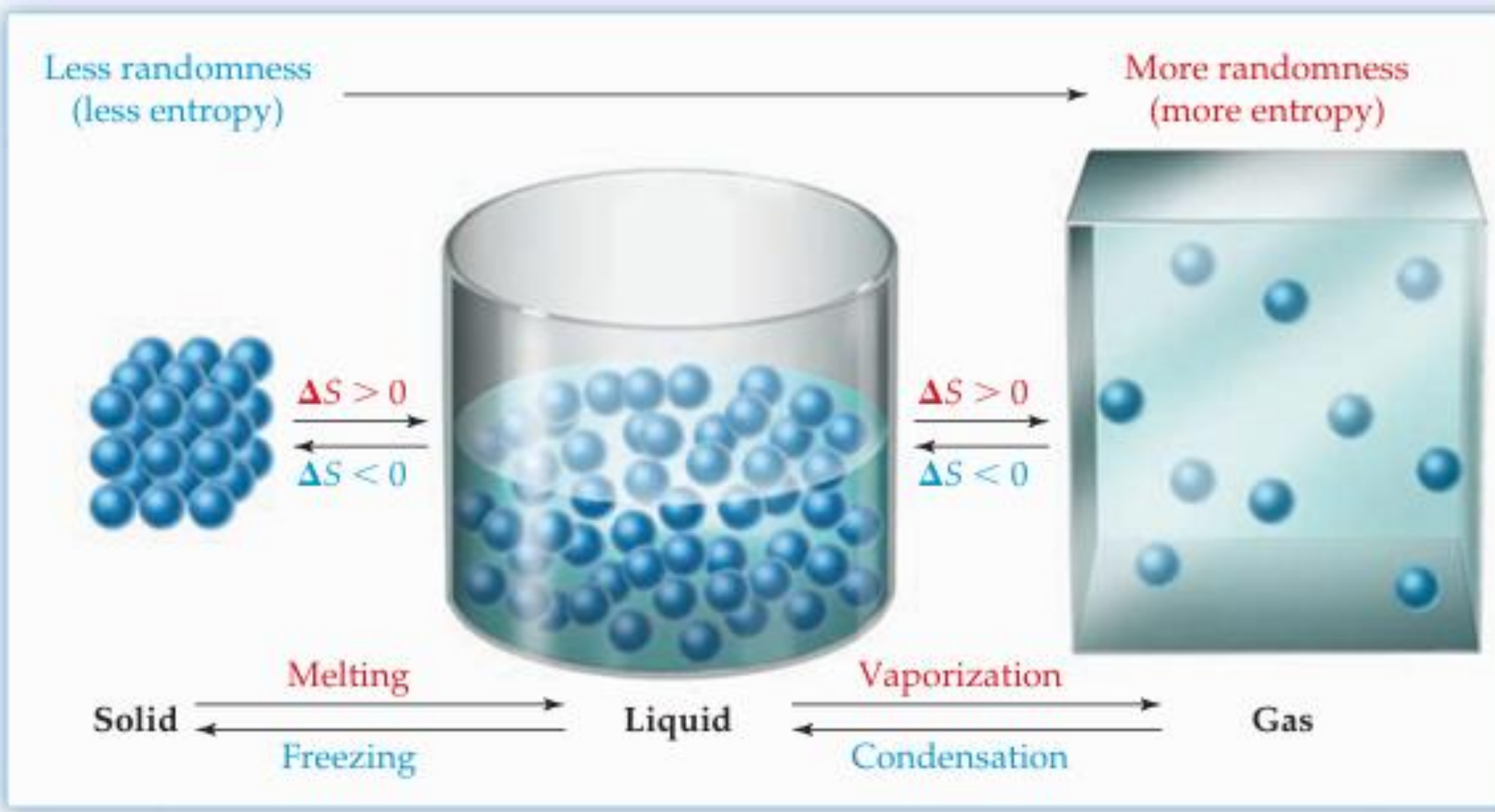
The reverse process, compression of the gas into a single bulb, is nonspontaneous.

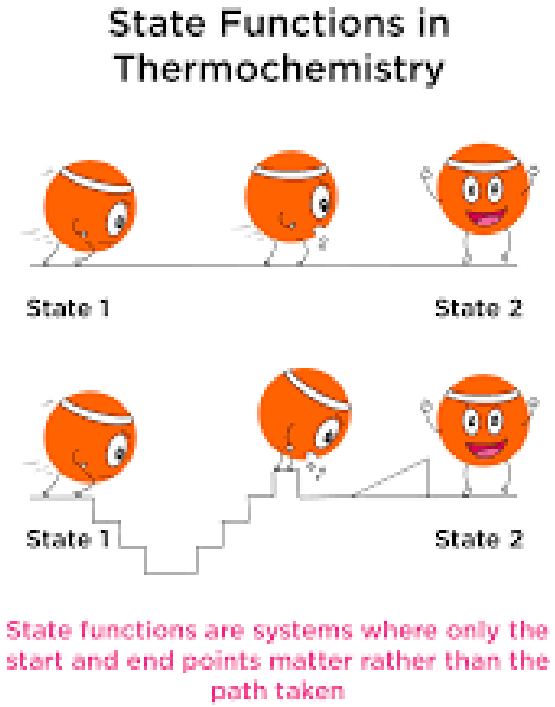
Entropy (ΔS) is defined as the measurement of the degree of randomness

It gives us knowledge about how things are developed and transformed (spontaneous or nonspontaneous)

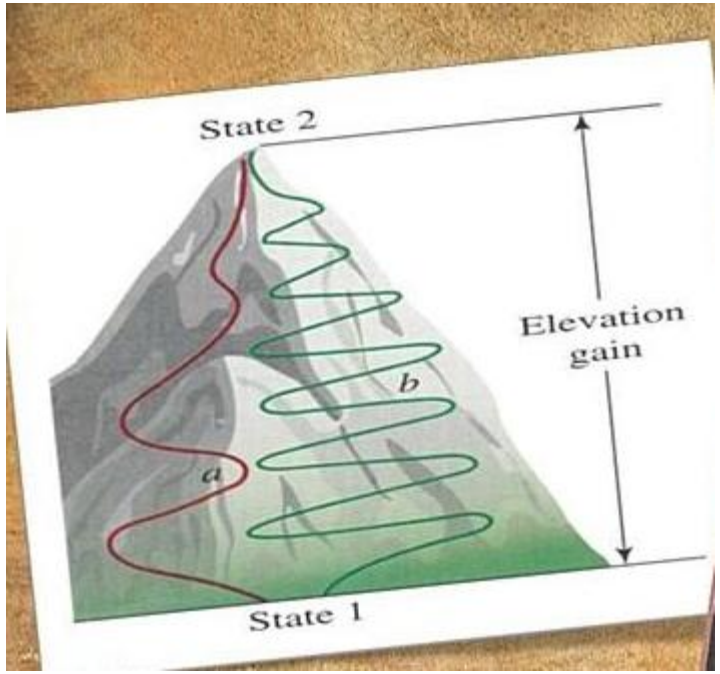


Entropy of the system **(ΔS positive > 0)** will always be **spontaneous.**


Less randomness
(less entropy)
More randomness
(more entropy)
ΔS > 0
ΔS < 0
ΔS > 0
ΔS < 0
Melting
Freezing
Vaporization
Condensation
Solid
Liquid
Gas



$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$



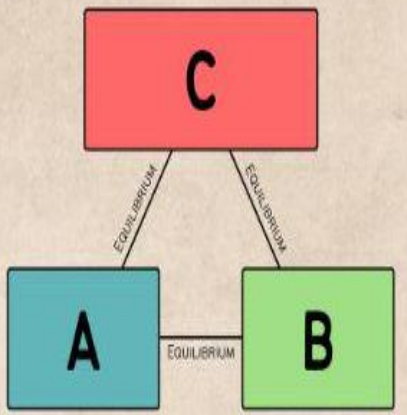
- Entropy is a state function
- State function or property whose value depends only on the present state (condition) of the system, not on the path used to arrive at that condition.
- Like; Pressure, volume, temperature

1.5. Thermodynamics laws

ZEROTH LAW
of Thermodynamics

Two systems in equilibrium with a third system are in thermal equilibrium with each other.

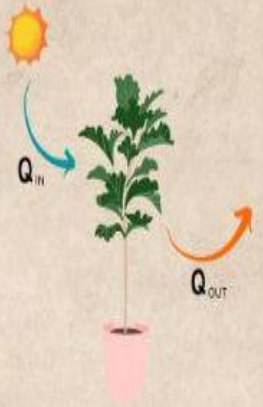
IF A = C AND B = C, THEN A = B



FIRST LAW
of Thermodynamics

Energy can neither be created nor destroyed; it can only be transferred from one form to another.

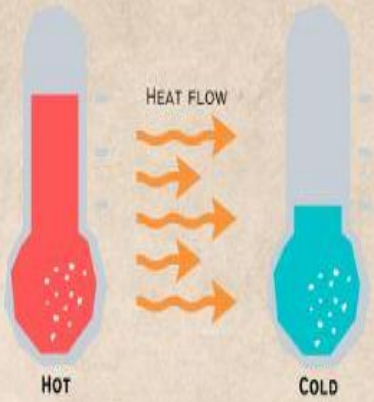
$\Delta U = Q - W$ OR $\Delta U = Q + W$



SECOND LAW
of Thermodynamics

The entropy of any isolated system always increases.

$\Delta S_{UNIV} > 0$



THIRD LAW
of Thermodynamics

The entropy of a system approaches a constant value as the temperature approaches absolute zero.

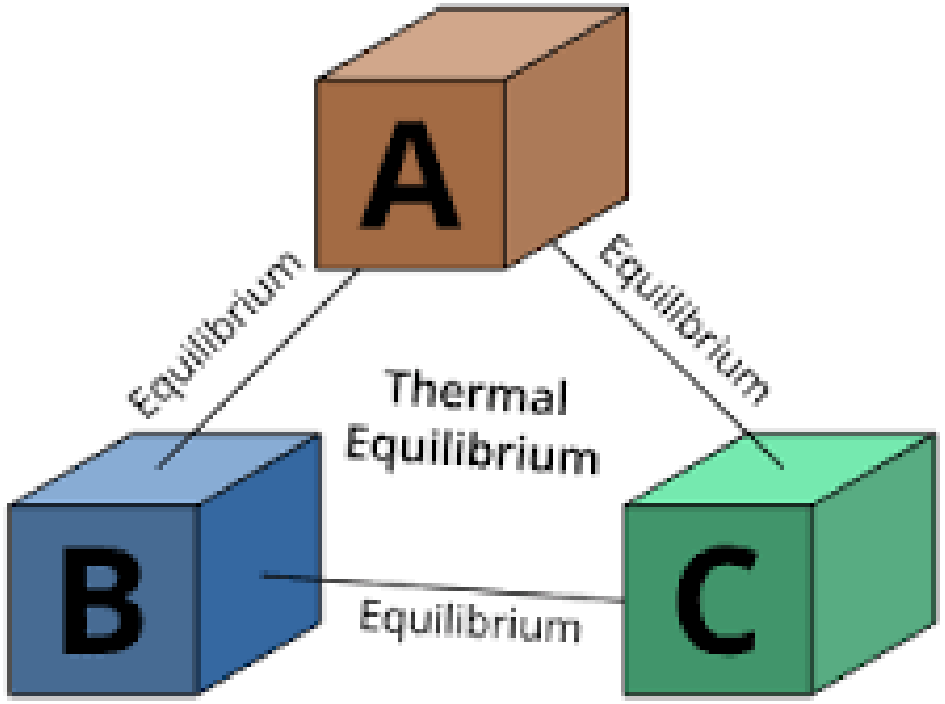
$S - S_0 = k_B \ln\Omega$

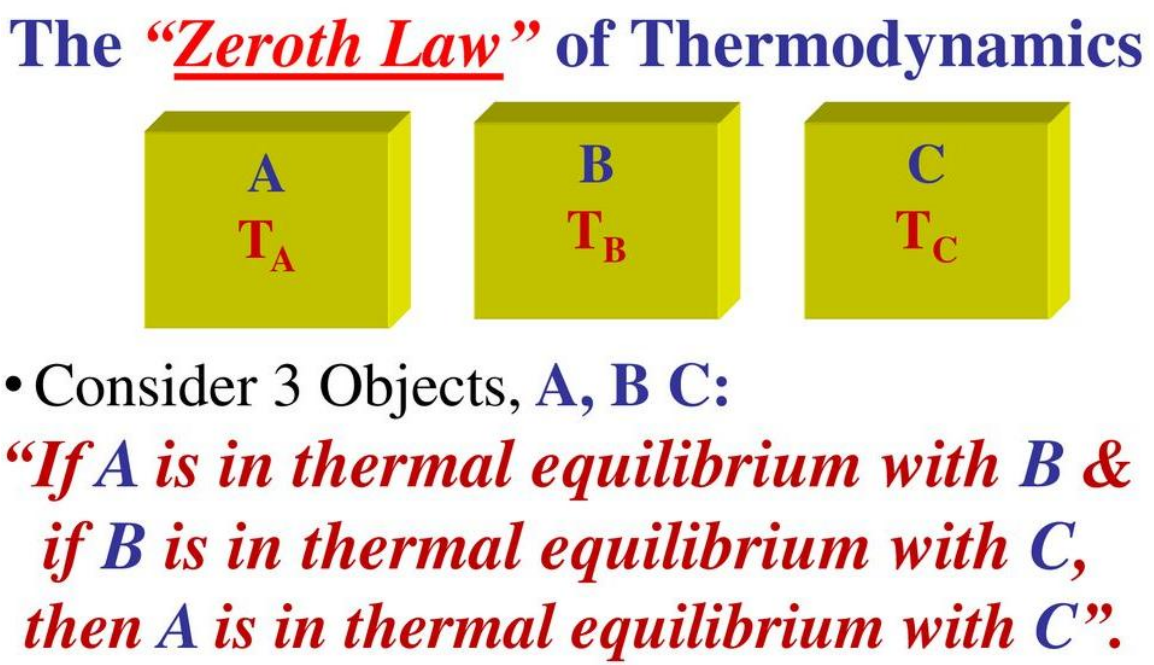


The zeroth law of thermodynamics

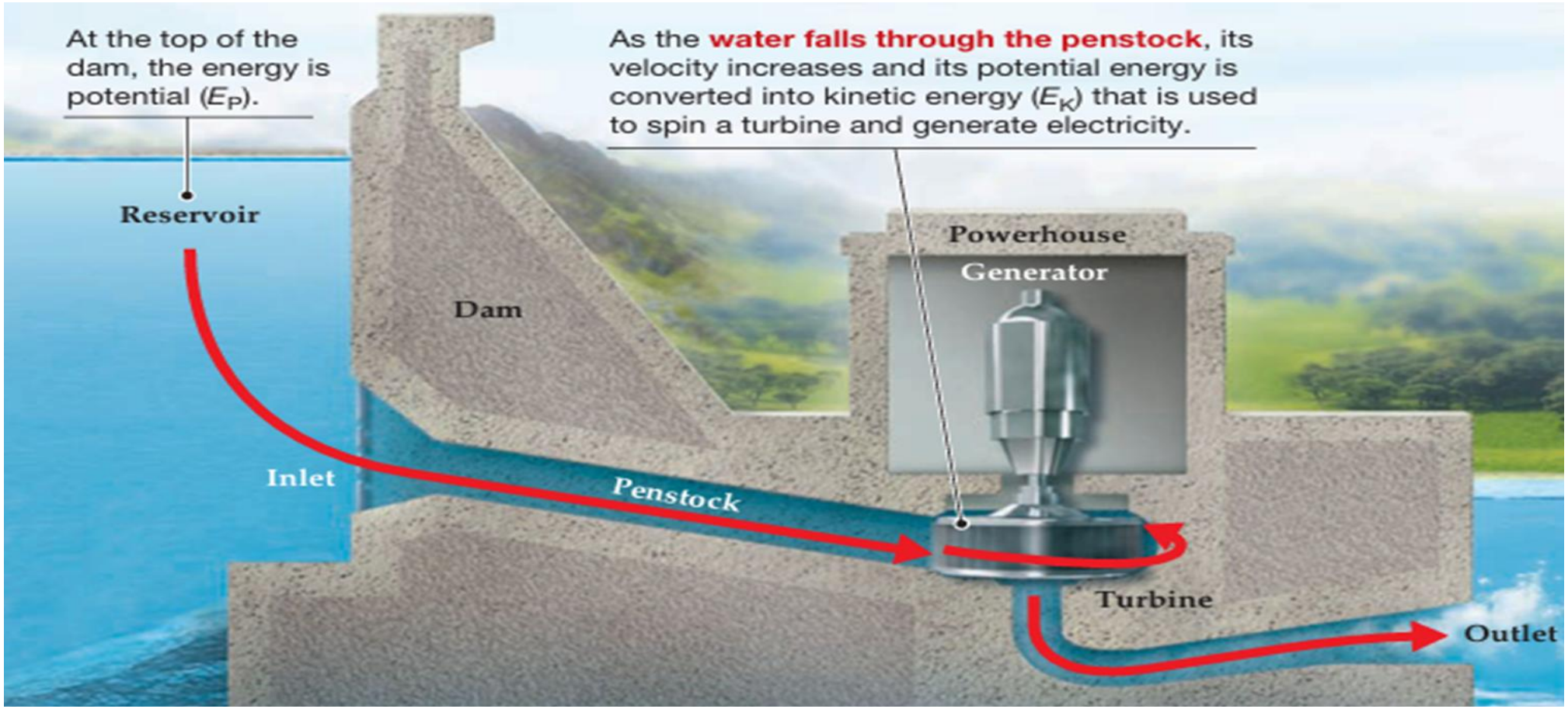
If two thermodynamic systems are both in thermal equilibrium with a third system, then the two systems are in thermal equilibrium with each other.

$$T_A = T_B = T_C$$





First Law of Thermodynamics (Conservation of Energy law)



Energy cannot be created or destroyed; it can only be converted from one form into another. the total amount of energy remains constant.

The change in internal energy of a system is equal to the heat added to the system minus the work done by the system.

$$\Delta U = Q - W$$

Change in internal energy	Heat added to the system	Work done by the system

Where:

Q is +ve if heat is added
Q is – ve if heat is released
w is +ve if work is done by the system
w is – ve if work is done on the system

Example1:

A gas is cooled, having 650 Joules of heat removed from it. During this time, 700 Joules of work is done on the gas. What is the change in the internal energy of the gas?

Solution:

$Q = -650J$

$W = -700J$

$\Delta U = ?$

$$\Delta U = Q - W$$

$$= (-650) - (-700) = 50J$$

Example 2:

A system receives 200 calories of heat and work done by the system is 736 J .What is change in its internal energy ?

Solution:

$$\Delta U = Q - W$$

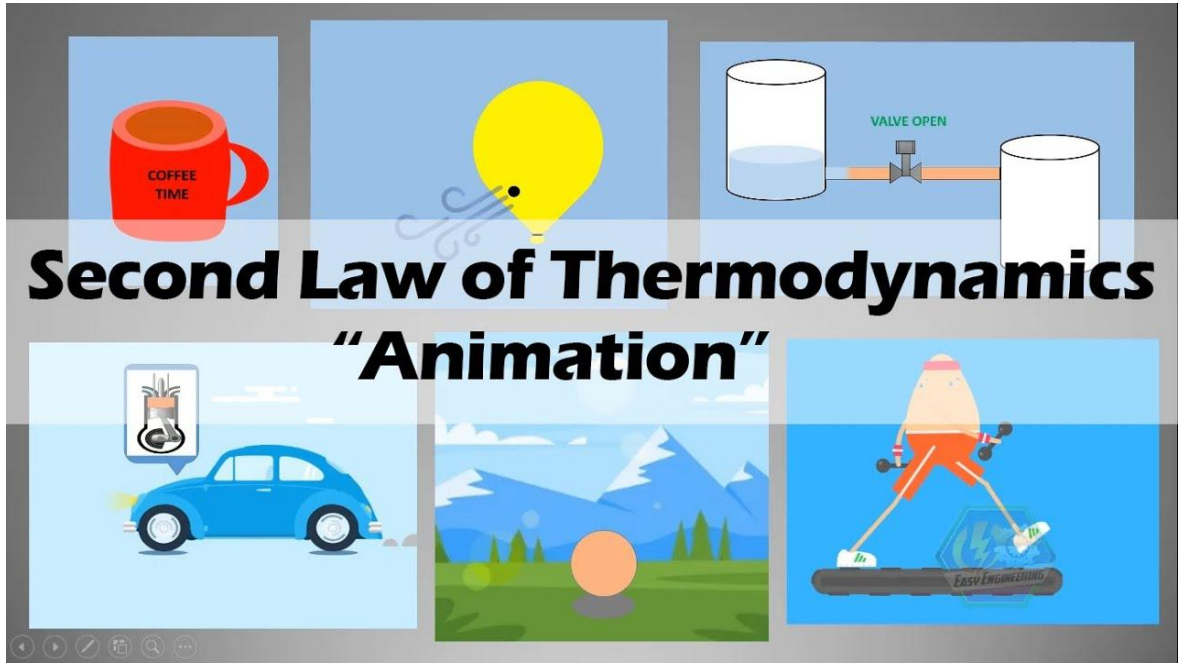
$$= (200)(4.186) - (736) = 101.2\,J$$

one calorie= 4.186 joules

Second law of thermodynamics

- A simple statement of the law is that heat always flows spontaneously from hotter to colder regions of matter.





- Processes that involve the transfer or conversion of heat energy are irreversible and always move toward more disorder.

Third law of thermodynamics

- The entropy (S) of a pure, perfectly crystalline substance approaches zero as its temperature approaches absolute zero (0 Kelvin)

$$S \rightarrow 0 \text{ as } T \rightarrow 0K$$

